We have seen that

$$\lim_{x\to a} c = c \tag{1}$$

$$\lim_{x\to a} x = a. \tag{2}$$

We can combine these limits with the Limit Laws to easily calculate many other limits.

Theorem. (The Limit Laws) Let $c \in \mathbb{R}$ be a constant and suppose the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist. Then

(i) $\lim_{x\to a}(f(x)+g(x)) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

(ii) $\lim_{x\to a}(cf(x)) = c\lim_{x\to a} f(x)$

(iii) $\lim_{x\to a}(f(x)g(x)) = \lim_{x\to a} f(x)\lim_{x\to a} g(x)$

(iv) $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$, provided that $\lim_{x\to a} g(x) \neq 0$

Example

$$\begin{aligned}
\lim_{x\to 1}(3x^2+2x+1) &= \lim_{x\to 1}(3x^2) + \lim_{x\to 1}(2x) + \lim_{x\to 1}(1) \qquad \text{(by (i))} \\
&= 3\lim_{x\to 1}(x^2) + 2\lim_{x\to 1}(x) + \lim_{x\to 1}(1) \qquad \text{(by (ii))} \\
&= 3\lim_{x\to 1}(x)\lim_{x\to 1}(x) + 2\lim_{x\to 1}(x) + \lim_{x\to 1}(1) \qquad \text{(by (iii))} \\
&= 3(1)(1) + 2(1) + 1 \qquad \text{(by (1) and (2))} \\
&= \boxed{6}.
\end{aligned}$$

1. Use the Limit Laws to calculate the limit $\lim_{x\to 2}(4x^3+6x^2+4x+4)$. Justify each step, as in the example above.

We can obtain new Limit Laws from (i)-(iv).

2. Show that (ii) and (i) imply that the limit of a difference is the difference of limits; i.e.,

$$\lim_{x\to a}(f(x)-g(x)) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x) \tag{3}$$

Hint: $-g(x) = (-1)\cdot g(x)$. Call this Limit Law (v).

3. Calculate the limit $\lim_{x\to 2}(x^2-3x+5)$ using the Limit Laws. Justify each step.

Here is another new Limit Law.

4. Use the Limit Laws to show that if n is a positive integer, then

$$\lim_{x\to a} x^n = \left(\lim_{x\to a} x\right)^n$$

Call this Limit Law (vi).

5. Let $p(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$ be any degree n polynomial. Use the Limit Laws to prove that, for any $a \in \mathbb{R}$,

$$\lim_{x \to a} p(x) = p(a);$$

that is,

$$\lim_{x \to a} (c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0) = c_n a^n + c_{n-1}a^{n-1} + \cdots + c_1 a + c_0.$$

This shows that the limit of any polynomial at any point $a \in \mathbb{R}$ may be evaluated by "direct substitution."

6. Use the Limit Laws to calculate $\lim_{x \to 2} \dfrac{x^2 - 3}{2x + 1}$. Justify each step.

7. Let $R(x) = \dfrac{p(x)}{q(x)}$ be any rational function (i.e., $p(x)$ and $q(x)$ are polynomials, with $q(x)$ not the constant polynomial $q(x) = 0$). Use the Limit Laws to prove that, for any point a in the domain of R (i.e., for any $a \in \mathbb{R}$ such that $q(a) \neq 0$). we have

$$\lim_{x \to a} R(x) = R(a);$$

that is,

$$\lim_{x \to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}.$$

This shows that the limit of any rational function at a point a in its domain may also be evaluated by direct substitution.

Additional Exercises

1. Given that

$$\lim_{x\to 2} f(x) = 4, \quad \lim_{x\to 2} g(x) = -2, \quad \lim_{x\to 2} h(x) = 0,$$

find the following limits, if they exist. If the limit does not exist, explain why.

(a) $\lim_{x\to 2}[f(x) + 5g(x)]$

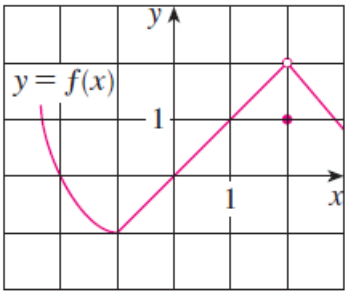
(b) $\lim_{x\to 2}[g(x)]^3$

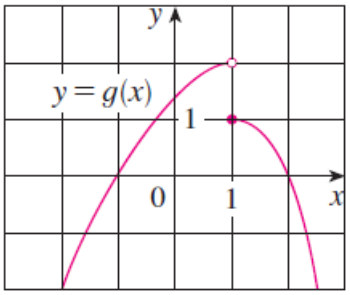
(c) $\lim_{x\to 2} \dfrac{3f(x)}{g(x)}$

(d) $\lim_{x\to 2} \dfrac{g(x)}{h(x)}$

(e) $\lim_{x\to 2} \dfrac{g(x)h(x)}{f(x)}$

2. The graphs of f and g are given. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.





(a) $\lim_{x\to 2}[f(x)+g(x)]$

(b) $\lim_{x\to 1}[f(x)+g(x)]$

(c) $\lim_{x\to 0}[f(x)g(x)]$

(d) $\lim_{x\to -1}[\dfrac{f(x)}{g(x)}]$

(e) $\lim_{x\to 2}[x^3 f(x)]$